Yichao Cai

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Research Statement

I study the theoretical and empirical foundations of multimodal representation learning, with a focus on contrastive learning geometry, cross-modal alignment, and causal/identifiable representations. My work combines information-theoretic analysis with controlled empirical evaluation to understand how language supervision shapes semantic structure in vision–language models, and how this perspective supports interpretable, reliable, and human-aligned AI systems.

Education

Ph.D. in Computer Science *Sep 2023 – Present*
University of Adelaide • Australian Institute for Machine Learning (AIML)
Advisor: Prof. Javen Qinfeng Shi

M.Sc. in Instrument Science and Technology *Sep 2016 – Jun 2019*
Wuhan University of Technology
Advisor: Prof. Xiao Zhou

B.Eng. in Measurement & Control Engineering *Sep 2012 – Jun 2016*
Wuhan University of Technology

Publications

CONFERENCE PAPERS

[1] **Yichao Cai***, Yuhang Liu*, Erdun Gao, Tianjiao Jiang, Zhen Zhang, Anton van den Hengel, Javen Qinfeng Shi.
On the Value of Cross-Modal Misalignment in Multimodal Representation Learning.
Advances in Neural Information Processing Systems (NeurIPS), 2025.
Spotlight (top ∼3% of all valid submissions) • *Equal contribution. [OpenReview] [PDF] [Code]

[2] **Yichao Cai**, Yuhang Liu, Zhen Zhang, Javen Qinfeng Shi.
CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts.
European Conference on Computer Vision (ECCV), 2024. [Springer] [PDF] [Code]

[3] Yuhang Liu, Dong Gong, **Yichao Cai**, Erdun Gao, Zhen Zhang, Biwei Huang, Mingming Gong, Anton van den Hengel, Javen Qinfeng Shi.
I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?
International Conference on Learning Representations (ICLR), 2026. [OpenReview] [PDF]

PREPRINTS

[1] **Yichao Cai**, Zhen Zhang, Yuhang Liu, Javen Qinfeng Shi.
The Geometric Mechanics of Contrastive Learning: Alignment Potentials, Entropic Dispersion, and Modality Gap.
arXiv preprint, 2026. [arXiv] [Code]

Research Interests

- **Vision–language models:** image–text representations for cross-modal retrieval and transfer; evaluation and benchmarking.
- **Contrastive & multi-view learning:** InfoNCE theory; geometry of representation spaces; training dynamics and failure modes.
- **Causal & identifiable representations:** concept learning; interpretable factors; language-guided disentanglement.
- **Robustness to data imperfections:** cross-modal misalignment; underspecification; practical mitigations.

Research Experience

Postgraduate Researcher *Sep 2023 – Present*
Australian Institute for Machine Learning (AIML), University of Adelaide
Advisor: Prof. Javen Qinfeng Shi

- Developed a theoretical framework for the geometry of InfoNCE-based contrastive learning, characterising alignment

potentials, entropic dispersion forces, and the emergence of modality gap.

- Studied cross-modal misalignment as a beneficial inductive bias in multimodal contrastive learning; established identifiability conditions and controlled benchmarks (NeurIPS 2025 Spotlight).
- Developed CLAP, a contrastive framework for content–style disentanglement via augmented language prompts (ECCV 2024).
- Design and execute empirical studies with rigorous ablations; curate controlled benchmarks; track experiments for reproducibility.

Visiting Student Researcher *May 2018 – Sep 2018*
California PATH, UC Berkeley
Supervisor: Dr. Ching-Yao Chan

- Conducted research on visual perception algorithms for autonomous driving.

Industry Experience

AI Engineer, Tellhow Software *Jun 2020 – Aug 2022*

- Built inspection and safety-analytics solutions for power-industry equipment using computer vision and automated analysis workflows; delivered end-to-end features from prototyping to production deployment.

Software Engineer, Huawei Technologies *Jul 2019 – Apr 2020*

- Developed and maintained communication service components; implemented production features and system integrations.

Teaching

Guest Lecturer, University of Adelaide *2025*
Courses: Statistical Machine Learning
Delivered lecture on recent advances in vision–language modeling. [Slides]

Head Tutor, University of Adelaide *2025*
Courses: Statistical Machine Learning

Teaching Assistant, University of Adelaide *2025 – 2026*
Courses: Neural Networks and Deep Learning • Using Machine Learning Tools • Concepts in AI and ML

Academic Service

Reviewer: Transactions on Machine Learning Research (TMLR) • International Conference on Learning Representations (ICLR) 2026 • International Conference on Machine Learning (ICML) 2026

Honors and Awards

NeurIPS 2025 Scholar Award, Neural Information Processing Foundation *Oct 2025*
University of Adelaide Research Scholarships *Sep 2023*
Award for Outstanding Graduates, Wuhan University of Technology *Jun 2019*